

# LUCAS KENJI MALHEIROS KAMIKAWA

Computer Engineering · Insper | Focus on Data Engineering & Data Science

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## PROFESSIONAL PROFILE

Computer Engineering student at Insper with a strong foundation in Python, SQL, and Data Science libraries. Hands-on experience in exploratory data analysis, predictive modeling (classification and regression), quantitative portfolio analysis, network analysis, ML pipelines, and RAG (Retrieval-Augmented Generation) architectures with LLMs. Participant in the Databricks Hackathon, where I built an end-to-end Churn prediction solution with a web interface accessible to managers. Disciplined and results-oriented, fluent in English, with strong ability to communicate technical insights to non-technical audiences.

## TECHNICAL SKILLS

**Languages:** Python, SQL (MySQL · SQLite), C, Java, TypeScript

**Libraries:** Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn, ChromaDB, Sentence-Transformers

**Cloud & Infra:** AWS (Lambda, API Gateway, DynamoDB, SQS, S3, IAM, CloudWatch), Docker

**Tools:** Git/GitHub, Linux, Databricks, FastAPI, Django REST

## FEATURED PROJECTS

### HRP/HERC — Quantitative Portfolio Allocation

- Implemented Hierarchical Risk Parity (HRP) and Hierarchical Equal Risk Contribution (HERC) strategies for portfolio allocation using historical market data.
- Built a backtesting pipeline comparing HRP, HERC, Equal-Weight, and Buy & Hold, evaluating performance via Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.
- Presented a critical analysis of both methodologies based on López de Prado (2016) and Raffinot (2018), including the trade-off between dendrogram structure and risk allocation.

### FinRAG — RAG for Financial Reports

- RAG pipeline for Q&A over financial documents — LLM + vector search over structured financial data.
- Dockerized full-stack application with separate frontend and backend containers, using Ollama for local LLM inference.

### CloudPay — Serverless Payments Platform on AWS

- Designed and implemented a 100% serverless payments platform MVP on AWS, with asynchronous processing via SQS, persistence in DynamoDB, and a React frontend hosted on S3.
- Architecture with 6 Lambda functions (Node.js), API Gateway HTTP API, and a decoupled producer/consumer flow with automatic retry.
- Ran load tests with Apache JMeter (100 concurrent users), identifying bottlenecks (DynamoDB Scan) and proposing optimizations (GSI, CloudFront, Cognito).

### Churn Prediction Interface | Databricks Hackathon

- Developed an end-to-end Churn prediction solution during the Databricks Hackathon, integrating an ML model with a web interface.
- Built an accessible frontend for non-technical managers, closing the gap between code and business analysis.

### Machine Learning – Adult Census

- Advanced EDA with missing data handling, encoding, and feature engineering on an income dataset.
- Predictive classification modeling with Scikit-learn; evaluated using performance metrics (accuracy, F1, ROC-AUC).
- Implemented linear and logistic regression, hypothesis testing, and analysis of qualitative and quantitative variables.

## EDUCATION & CERTIFICATIONS

- B.Eng. in Computer Engineering — Insper (2023 – 2028)
- Google Cybersecurity Professional Certificate — security fundamentals, networking, Linux, SQL, and automation with Python.
- AWS Academy Graduate — Cloud Foundations — AWS Cloud Computing fundamentals.

## LANGUAGES

- English: Advanced/Fluent (C1) — 4 years living in the U.S. + exchange program in Canada.
- Portuguese: Native.
- German: Basic/Intermediate (B1).